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Integrated Curricula and Preparation for College Mathematics

In the early 1990s, after a long series of disappointing results on national and international mathematics achievement tests—for example, TIMSS (1998) and NAEP (Campbell, Hombo, and Mazzeo 2000)—the National Science Foundation (NSF) funded the development of thirteen complete mathematics programs at the elementary school, middle school, and secondary school levels.

By NSF edict, each reform curriculum had to align with the basic process principles and content specifications articulated in NCTM's *Curriculum and Evaluation Standards for School Mathematics* (1989). At the secondary school level, these NSF-funded curricula, released between 1997 and 1999, extended the mathematical domains addressed each year by including attention to algebra, geometry, probability, statistics, and topics in discrete mathematics. The pedagogical guidelines were also more expansive and quite different from the methodology commonly used at the time. As a result, implementing these materials posed a different set of challenges for teachers.

Many teachers were quite happy with the new curricula and impressed with the levels of student involvement and accomplishment. However, some parents and mathematicians were opposed to them, referring to the new curricula as "fuzzy mathematics." In response, in 2007 NSF funded the Minnesota Mathematics Achievement Project (MNMAP) to explore the empirical relationship between high school mathematics curricula and college-level mathematics performance.

MNMAP is a longitudinal program of research. Various subsets of its database of more than 16,000 students from 366 high schools who attended thirty-two four-year regional colleges and universities in the upper Midwest were involved in the MNMAP studies. All students graduated from a Minnesota high school and completed three or more years of college-intending high school mathematics. This group presented a unique opportunity to examine the mathematics achievement of students who had recently completed one of two distinctively different types of high school curricula—one NSF-funded and one commercially developed. These students have now graduated from regional four-year colleges and universities.

We will summarize five of these MNMAP studies, each of which used quantitative methodology. All studies were concerned with the efficacy of three first-edition secondary school curricula—Contemporary Mathematics in Context (Core Plus), Interactive Mathematics Project (IMP), and Mathematics Modeling Our World (MMOW)—as measured by student performance in college mathematics. Each program received an additional grant from NSF to prepare second-edition materials.

TWO VIEWS OF HIGH SCHOOL MATHEMATICS CURRICULA

The three NSF-funded curricula considered here offer students opportunities to communicate and reason mathematically in ways not found in traditional textbooks, referred to here as commercially developed curricula. In the NSF-funded curricula,

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problems are introduced within some mathematically rich, real-world context. Rarely are raw numbers or symbols manipulated simply for practice.

Two criticisms of NSF-funded curricula were that they did not pay enough attention to algorithmic development and that more extensive practice needed to be included in the lessons. The cognitive perspective underlying NSF-funded curricula expands the nature and extent of mathematical and instructional goals. This means that algorithmic development and maintenance is one of several important goals but, because the content has been expanded, a lesser portion of classroom time is available for the development and maintenance of skills and procedures. Critics believed that this approach would leave students unprepared for college-level mathematics. This hypothesis is unsupported by empirical evidence.

Method

In the past, the school mathematics curriculum has not received the research attention that it deserves. Rarely have two curricula been contrasted to see which has been more effective over the long run. The NSF-funded curricula are probably the most researched curricula in our nation's history. Previous studies assessed topic-specific achievement, calculus 1, problem-solving ability, attitude, and tenacity, but we identified no studies that used large databases from multiple postsecondary institutions to follow students through four full years of college coursework. The MNMAP studies discussed here are an attempt to fill this void and provide insight into the relationship between high school mathematics curricula and college-level performance.

In educational settings, units of analysis—students, schools, institutions—are often embedded, or nested, within larger units. For example, students are nested within classrooms that are nested within schools. Units within a larger unit tend to be more similar to one another and are therefore not independent (e.g., students in a classroom tend to be more similar to one another than they are to students in a different classroom). Traditional analyses, such as multiple regression, does not take this dependence into account, but hierarchical linear modeling, the primary analytic tool used in the MNMAP studies, does, thus significantly improving our ability to understand relationships among the many meaningful and relevant variables that typically are present in educational studies.

The first four MNMAP studies discussed here compared the achievement and course-taking patterns of students who had completed three, four, or five years of an NSF-funded curriculum with students who had used a commercially developed program or the University of Chicago School Mathematics Project. The fifth study relates to the likelihood of students completing a college STEM major—engineering or mathematics, in particular—as a function of the type of their high school mathematics curriculum.

New college students in 2002 and 2003 were some of the first students to have completed three or four years of one of the NSF-funded curricula. Some also completed a fifth year of secondary school mathematics (usually calculus), neither a non-NSF-funded nor a noncommercially developed course. Students in the latter cohort used materials published by one of approximately twenty programs, including those of Laidlaw, Holt, Addison Wesley, and Saxon. Students in the NSF-funded cohort were enrolled in courses using either Core Plus, IMP, or MMOW. The MNMAP website (www.cehd.umn.edu/ci/mnmap/) contains additional information about each study discussed here. In the three longitudinal studies, MNMAP examined all students' mathematics coursework taken over eight semesters of college work.

The MNMAP studies discussed here are an attempt to provide insight into the relationship between high school mathematics curricula and college-level performance.

Independent Variables

Student level variables (independent variables or covariates) were (1) high school curricula compared — Variable of major interest in each of these studies: (1a) NSF-funded vs. commercially developed curricula; (1b) Core Plus vs. commercially developed curricula, (1c) NSF-funded vs. commercially developed and UCSMP; (2) ACT scores; (3) number of years of high school mathematics (three, four, or five); (4) high school mathematics GPA; (5) high school percentile rank; (6) college major; (7) gender; (8) ethnicity; and (9) basic skills state test scores.

Dependent Variables

The dependent variables considered included these: (a) the difficulty level of the first college mathematics course taken (developmental, college algebra, calculus, etc.); (b) the difficulty level of the last college mathematics course taken (developmental through upper-division mathematics courses); (c) the grades or patterns of grades that students earned in mathematics courses (achievement); (d) the number of mathematics courses completed (persistence); (e) the pattern of difficulty levels of all mathematics courses completed over multiple semesters; (f) the likelihood of completing a STEM

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Table 1 Summary of the Design and Participants of the Five MNMAP Studies

Study	Reference	Participants	Independent (I) and Dependent (D) Variables	Design
1	Harwell, Post, et al. (2009)	$N = 1{,}667$; 85 high schools	I: (1a), (2), (3), (4), (5), (8), (9) D: (a), (c)	Retrospective quasi-experimental, cross-sectional student data from a single institution at one point in time
2	Harwell, Medhanie, et al. (2011)	$N = 1{,}588$; 209 high schools	I: (1b), (2), (3), (4), (6), (7), (8) D: (a), (b), (c), (e), (g)	Retrospective, longitudinal analysis (four years of college coursework); data from a single institution
3	Post et al. (2010)	$N = 4{,}144$; 266 high schools	I: (1c), (2), (3), (4), (6), (7), (8) D: (a), (b), (c), (d), (e)	Retrospective, longitudinal analysis; data from a single institution
4	Harwell, Post, et al. (in press)	$N = 10{,}190$; 266 high schools; 32 four-year postsecondary institutions	I: (1c), (2), (3), (4), (6), (7), (8) D: (a), (b), (c), (e), (g)	Retrospective, longitudinal analysis; data from multiple institutions
5	LeBeau et al. (2012)	$N = 3{,}459$; 1,166 STEM majors; 537 completed an engineering or mathematics degree	I: (1c), (2), (3), (5), (7), (8) D: (f), (h)	Retrospective, cross-sectional analysis; data from a single institution

major in college; (g) the student's grade in a first calculus course; and (h) within STEM majors, the likelihood of completing an engineering or a mathematics major. This research was conducted initially with students from one large university and later with students from thirty-two four-year regional institutions varying in size, selectivity, mission, and type (public or private). The key findings are consistent over the five studies and across the thirty-two institutions.

See **table 1** for a summary of the design and participants for each study. Note that the references to independent and dependent variables refer to the lists above. **Table 2** provides a summary of the results across the first four studies.

SUMMARY OF RESULTS

Study 1: Harwell, Post, et al. (2009)

When we examined the entire NSF-funded cohort, we found differences in the difficulty level of initial course enrollment favoring students in commercially developed programs. These enrollments, from both types of curricula, included developmental (non-credit-bearing) and nondevelopmental courses (college algebra, calculus 1 and 2, and higher-level courses). There were no curricular-related differences in the grades attained in these courses.

Study 2: Harwell, Medhanie, et al. (2011)

When we examined only students who took two or more college mathematics courses, the curricular-related differences mentioned in study 1 disappeared. That is, there were no differences in the difficulty level or achievement (grades) of the initial course enrollment. Further, there were no differences in the longitudinal difficulty levels of coursework completed or in the grades attained in those courses examined over eight semesters of coursework. We also found no curricular-related differences in the number of mathematics courses completed.

Study 3: Post et al. (2010)

When we partitioned students according to their previous achievement scores (forming three groups), the middle group (those with ACT scores of 22 to 26) and the high group (those with ACT scores higher than 26) showed no significant differences in any of the dependent variables (the difficulty levels of the initial and final courses, achievement, and persistence) that we examined. For the group of students with the lowest ACT scores, we found a difference in the difficulty level of initial course enrollment (primarily developmental) favoring the cohort using commercially developed materials, with no curricular differences in grades earned.

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Table 2 MNMAP Summary of Results for Studies 1–4

	Number of Students	Difficulty Level of Initial College Mathematics Course	Grade of Initial College Mathematics Course	Persistence (Number of College Mathematics Courses)	Longitudinal Grade Trajectories	Longitudinal Difficulty Trajectories
Study 1	1,667	CD > NSFF	NSD			
Study 2	1,704	NSD	NSD	NSD	NSD	NSD
Study 3 (ACT strata)	4,144	Low 1/3: CD > NSFF Middle and Top 1/3: NSD	NSD	NSD	NSD	NSD
Study 4 (Multi-institutional)	10,300	CD > NSFF	NSD (last course)		NSD	NSD

Note: NSFF = National Science Foundation funded; CD = Commercially developed curricula; NSD = No significant difference

Study 4: Harwell, Post, et al. (in press)

When we expanded the sample to thirty-two four-year regional institutions and more than 10,000 students, the results were consistent. When these students were compared with students in the cohort using commercially developed materials, the only difference was in relation to the difficulty level of the first college mathematics course. That is, students in the NSF-funded cohort, regardless of institutional parameters, were as well prepared for college mathematics as their counterparts who used commercially developed materials, suggesting that earlier results related to preparing students equally well for college mathematics are generalizable. Again, students in the NSF-funded cohort enrolled initially in lower-level coursework but achieved equally well in that course and in subsequent courses. We found no differences at any of the difficulty levels of the last college mathematics course completed.

Study 5: LeBeau et al. (2012)

When we restricted our sample to only those students who had completed a STEM major and then to students who had completed an engineering or mathematics major, the results were again consistent. That is, we found no relationship between students' high school curriculum and their likelihood of completing an engineering or a mathematics major in college.

EXPLANATION OF RESULTS

Teachers will notice that we used the terms "equally well prepared," "no differences," and "no curricular relationship" in our summary. What exactly are the implications of these phrases, and why are these implications important for classroom teachers? These results suggest that multiple, very different approaches can prepare high school students equally well for college mathematics.

To qualify for funding, all NSF-funded curricula must have adhered to NCTM's *Curriculum and Evaluation Standards for School Mathematics* (1989). It follows that each of these curricula reflect in various ways the "best practices" in our field. Although NCTM's 1989 publication and its *Principles and Standards for School Mathematics* (NCTM 2000) are not exclusively research based, they do reflect many research implications that have accumulated over the past several decades relating to student learning and conceptual development, effective teaching, the sociology of the classroom, uses of technology, and appropriate assessment practices. Classrooms based on these principles portray a very different learning environment—one that is student centered. In such classrooms, the teacher's role changes from primarily that of a lecturer to that of a coach who promotes both student independence and interdependence. This new role routinely highlights the applications of mathematics in the real world and preempts the ever-present student question, "Why do I have to learn this?"

Does such a student-oriented environment become a disadvantage for students later on? Our results suggest that it does not. In fact, these studies suggest that, aside from curricular-related differences in NSF-funded and commercially developed curricula in the difficulty level of the first college mathematics course, primarily for students enrolling in lower-level courses, we found no other differences in any of the dependent variables.

As we attempted to explain these results further, we focused on the characteristics of NSF-funded curricula. Each of the three curricula—Core Plus, IMP, and MMOW—was designed to appeal to a wide spectrum of the high school population. Key elements include the integration of five mathematical domains every year, widespread use of contextually laden problem situations, group-oriented problem solving, and use of a wide variety of

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technology, including spreadsheets, graphing calculators, computer algebra systems (CAS), and applications such as The Geometer's Sketchpad® and Fathom Dynamic Data™ software.

These elements engage students in the conceptually oriented aspects of mathematics and its applications while necessarily limiting the time devoted to routine paper-and-pencil manipulations. This approach represents a substantial departure from the status quo and consequently has contributed to the acrimonious interchanges alluded to earlier.

The NSF-funded curricula included in these studies do prepare students for college mathematics equally as well as commercially developed approaches.

The problem-solving aspects of NSF-funded curricula engage students in problems whose solutions often consume several days of class time. This more accurately depicts the interdisciplinary nature of problem solving as it occurs in the real world. The tenacity acquired from these elaborate problem-solving experiences apparently served students well as they pursued college-level mathematics.

These characteristics do not appear only in the NSF-funded curricula. However, they are found there in a more coordinated fashion and in more abundance than in more traditional approaches.

A LOOK AHEAD

The Common Core State Standards for Mathematics (CCSSI 2010), designed to establish national standards for school mathematics, has been endorsed by more than forty states and the District of Columbia. The intent is "the specification of content that all students must study in order to be college and career ready" (CCSSI 2010, p. 4).

CCSSM has developed two major model course pathways—a traditional one and and an integrated one. Specific mathematics content by grade or course level is suggested within each of the specified pathways, including a transpathway perspective of an expanded view of the various mathematical content domains to be addressed each year. The NSF-funded curricular approaches, by virtue of addressing five mathematical domains each year and selecting problems from a wide variety of disciplines, has had an integrated perspective from the beginning.

CCSSM also endorses a series of mathematical practices that relate to NCTM's original five Process Standards: Problem Solving, Reasoning and Proof, Communication, Connections, and Representation. The Common Core Standards for Mathematical Practice are (1) make sense of problems and persevere in solving them; (2) reason abstractly and quantitatively; (3) construct viable arguments and critique the reasoning of others; (4) model with mathematics; (5) use appropriate tools strategically; (6) attend to precision; (7) look for and make use of structure; and (8) look for and express regularity in repeated reasoning (CCSSI 2010, pp. 6–8).

These practices are subtle yet profound, and implementation in the manner intended will represent a substantial departure in both content and method from the status quo. These core practices are not unrelated to more traditional approaches to high school mathematics; rather, these practices are more ably reflected in the NSF-funded approaches. This should not be surprising—these curricula have reflected alternative types of classroom mathematical activities for more than a decade.

SUMMARY AND IMPLICATIONS

Newer approaches to school curricula are always adapted reluctantly, especially so in mathematics. After all, everyone is interested in providing the best possible experiences for his or her own students. In one local suburban district, NSF-funded curricula were implemented for grades K–12 for all students more than a decade ago. Even though this district was high achieving before the change, it is now one of the top two or three highest-performing districts in the state regarding graduation rates, standarized test scores, the number and percentage of students in AP calculus and AP statistics courses, the percentage of students who take four full years of college-intending mathematics (through Core Plus IV), and the percentage of students enrolled in four-year postsecondary institutions (Monson 2011).

Earlier concerns about NSF-funded curricula—primarily that they will not prepare students for college-level mathematics in general and calculus in particular—have not been validated by these studies. The NSF-funded curricula included in these studies do prepare students for college mathematics equally as well as commercially developed approaches. Further, there may also be legitimate precollege pedagogical and sociological reasons for adopting NSF-funded curricula.

For schools and teachers interested in materials that have a decidedly different perspective, that appeal to a wider audience, and that promote a higher degree of student engagement, the NSF-funded curricula are a viable option.

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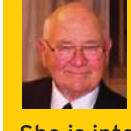
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